Forensics CTF Writeup - THM - DFIR

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Objective

> You will get a memory dump of a compromised system, do some digital forensics to explore the inside

Tools Used

- Volatility 2 - Volatility Framework
- Strings
- Grep

Task 1 - Volatility Forensics

Question 1 - What is the Operating System of this Dump file? (OS name)

```
python2.7 ~/volatility/vol.py -f victim.raw imageinfo
```

What this command does

This command uses Volatility Framework to analyze the memory dump file (`victim.raw`) and identify important system information such as:

- Operating system version
- Architecture (32-bit or 64-bit)
- Suggested memory profile
- Kernel details
- Timestamp of the captured memory

The `imageinfo` plugin is usually the **first command** used in memory forensics because many other Volatility plugins require the correct profile to work

properly.

> Answer - windows

```
┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 1]
└─$ python2.7 ~/volatility/vol.py -f victim.raw imageinfo
Volatility Foundation Volatility Framework 2.6.1
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_24000, Win2008R2SP1x64_23418, Win2008R2SP1x64,
 Win7SP1x64_24000, Win7SP1x64_23418
                     AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
                     AS Layer2 : FileAddressSpace (/home/kav3nirmal141/Tryhackme/Forensics /Task 1/victim.raw)
                      PAE type : No PAE
                           DTB : 0x187000L
                          KDBG : 0xf800028420a0L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0xfffff80002843d00L
             KUSER_SHARED_DATA : 0xfffff78000000000L
           Image date and time : 2019-05-02 18:11:45 UTC+0000
     Image local date and time : 2019-05-02 11:11:45 -0700
```

Question 2 - What is the PID of SearchIndexer?

```
python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 pslist
```

What this command does

The `pslist` plugin in Volatility Framework displays the list of active processes found in the memory dump.

This helps investigators:

- Identify suspicious or malicious processes
- View parent-child process relationships
- Analyze process creation times
- Detect unusual executables running in memory

The `--profile=Win7SP1x64` argument tells Volatility how to properly interpret Windows kernel structures inside the dump.

```
0xfffffa8002a1bb30 svchost.exe          752    504      7      235      0      0 2019-05-02 18:02:51 UTC+0000
0xfffffa8002d70650 svchost.exe          852    504     22      473      0      0 2019-05-02 18:02:51 UTC+0000
0xfffffa8002d9c780 svchost.exe          892    504     17      427      0      0 2019-05-02 18:02:51 UTC+0000
0xfffffa8002dbe9e0 svchost.exe          920    504     29      878      0      0 2019-05-02 18:02:51 UTC+0000
0xfffffa8002e3db30 svchost.exe          400    504     10      281      0      0 2019-05-02 18:02:56 UTC+0000
0xfffffa8002e57890 svchost.exe         1004    504     20      379      0      0 2019-05-02 18:02:56 UTC+0000
0xfffffa8002dfdab0 spoolsv.exe         1140    504     12      279      0      0 2019-05-02 18:02:57 UTC+0000
0xfffffa8002f2cb30 svchost.exe         1268    504     17      297      0      0 2019-05-02 18:02:59 UTC+0000
0xfffffa8002f81460 svchost.exe         1368    504     20      295      0      0 2019-05-02 18:02:59 UTC+0000
0xfffffa8003148b30 taskhost.exe        1788    504      8      159      1      0 2019-05-02 18:03:09 UTC+0000
0xfffffa8003172b30 explorer.exe        1860   1756     19      645      1      0 2019-05-02 18:03:09 UTC+0000
0xfffffa800315eb30 dwm.exe             1896    892      3       69      1      0 2019-05-02 18:03:09 UTC+0000
0xfffffa800300d700 VBoxTray.exe        1600   1860     13      141      1      0 2019-05-02 18:03:25 UTC+0000
0xfffffa8003367060 SearchIndexer.      2180    504     11      629      0      0 2019-05-02 18:03:32 UTC+0000
0xfffffa80033f6060 WmiPrvSE.exe        2876    628      5      113      0      0 2019-05-02 18:03:55 UTC+0000
0xfffffa8003162060 svchost.exe         1820    504     11      317      0      0 2019-05-02 18:05:09 UTC+0000
0xfffffa8003371540 wmpnetwk.exe        2464    504     14      440      0      0 2019-05-02 18:05:10 UTC+0000
0xfffffa80014eeb30 taskhost.exe        1148    504      8      176      0      0 2019-05-02 18:09:58 UTC+0000
```

> Answer - 2180

Question 3 - What is the last directory accessed by the user?

```
python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 shellbags
```

What this command does

The `shellbags` plugin in Volatility Framework extracts Windows ShellBag information from memory.

ShellBags are Windows registry artifacts that store:

- Folder access history
- Explorer navigation activity
- Removable drive interaction
- User browsing behavior inside Windows Explorer

This helps investigators understand:

- What folders the user accessed

- Whether external drives were connected
- Possible attacker activity on the system

```
0       0     Microsoft      1970-01-01 00:00:00 UTC+0000   1970-01-01 00:00:00 UTC+0000   1970-01-01 00:00:00 UTC+0000   DIR                    C:\P
**************************************************************************

**************************************************************************
Registry: \??\C:\Users\victim\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\1\1\2
Last updated: 2019-04-27 10:38:03 UTC+0000
Value   Mru   File Name      Modified Date                  Create Date                    Access Date                    File Attr              Path
------  ----  -------------  -----------------------------  -----------------------------  -----------------------------  ---------------------  ----
0       0     Capture        2019-04-27 10:36:06 UTC+0000   2019-04-18 00:49:00 UTC+0000   2019-04-27 10:36:06 UTC+0000   DIR                    C:\P
**************************************************************************

**************************************************************************
Registry: \??\C:\Users\victim\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\1\2\0
Last updated: 2019-04-27 10:48:33 UTC+0000
Value   Mru   File Name      Modified Date                  Create Date                    Access Date                    File Attr              Path
------  ----  -------------  -----------------------------  -----------------------------  -----------------------------  ---------------------  ----
0       0     deleted_files  2019-04-27 10:30:26 UTC+0000   2019-04-27 10:38:24 UTC+0000   2019-04-27 10:38:24 UTC+0000   NI, DIR                Z:\l
**************************************************************************

**************************************************************************
Registry: \??\C:\Users\victim\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\5\2\2
Last updated: 2019-04-23 06:14:25 UTC+0000
Value   Mru   File Name      Modified Date                  Create Date                    Access Date                    File Attr              Path
------  ----  -------------  -----------------------------  -----------------------------  -----------------------------  ---------------------  ----
1       0     NEWFOL~1       2019-04-23 06:14:26 UTC+0000   2019-04-23 06:14:26 UTC+0000   2019-04-23 06:14:26 UTC+0000   DIR                    Life
0       1     VIRUSS~1       2019-04-18 01:17:14 UTC+0000   2019-04-18 01:17:14 UTC+0000   2019-04-18 01:17:14 UTC+0000   DIR                    Life
5c632fe8ab2727adc9ac7b1b59c3be8
**************************************************************************
```

> Answer - deleted_files

Task 2

Question 1 - There are many suspicious open ports; which one is it? (ANSWER format: protocol:port)

```
python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 netscan
```

What this command does

The `netscan` plugin in Volatility Framework scans memory for active and recently used network connections.

This plugin helps investigators identify:

- Open TCP/UDP ports
- Listening services
- Suspicious communications
- Malware command-and-control activity

- Which process owns each connection

It is one of the most important plugins during malware investigations.

```
┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 1]
└─$ python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 netscan
Volatility Foundation Volatility Framework 2.6.1
Offset(P)          Proto    Local Address                  Foreign Address      State            Pid      Owner          Created
0x5c201ca0         UDPv4    0.0.0.0:5005                   *:*                                   2464     wmpnetwk.exe   2019-05-02 18:05:14 UTC+0000
0x5c201ca0         UDPv6    :::5005                        *:*                                   2464     wmpnetwk.exe   2019-05-02 18:05:14 UTC+0000
0x5c49cbb0         UDPv4    0.0.0.0:59471                  *:*                                   1368     svchost.exe    2019-05-02 18:03:06 UTC+0000
0x5c4a31c0         UDPv4    0.0.0.0:59472                  *:*                                   1368     svchost.exe    2019-05-02 18:03:06 UTC+0000
0x5c4a31c0         UDPv6    :::59472                       *:*                                   1368     svchost.exe    2019-05-02 18:03:06 UTC+0000
0x5c4ac630         UDPv4    0.0.0.0:3702                   *:*                                   1368     svchost.exe    2019-05-02 18:03:14 UTC+0000
0x5c4ac630         UDPv6    :::3702                        *:*                                   1368     svchost.exe    2019-05-02 18:03:14 UTC+0000
0x5c519b30         UDPv4    0.0.0.0:3702                   *:*                                   1368     svchost.exe    2019-05-02 18:03:14 UTC+0000
0x5c537ec0         UDPv4    0.0.0.0:3702                   *:*                                   1368     svchost.exe    2019-05-02 18:03:14 UTC+0000
0x5c690360         UDPv4    0.0.0.0:0                      *:*                                   1004     svchost.exe    2019-05-02 18:02:56 UTC+0000
0x5c690360         UDPv6    :::0                           *:*                                   1004     svchost.exe    2019-05-02 18:02:56 UTC+0000
0x5c6918e0         UDPv4    0.0.0.0:5355                   *:*                                   1004     svchost.exe    2019-05-02 18:02:56 UTC+0000
0x5c6918e0         UDPv6    :::5355                        *:*                                   1004     svchost.exe    2019-05-02 18:02:56 UTC+0000
0x5c692940         UDPv4    0.0.0.0:5005                   *:*                                   2464     wmpnetwk.exe   2019-05-02 18:05:14 UTC+0000
0x5c692ae0         UDPv4    0.0.0.0:5355                   *:*                                   1004     svchost.exe    2019-05-02 18:02:56 UTC+0000
0x5c7bac70         UDPv4    0.0.0.0:5004                   *:*                                   2464     wmpnetwk.exe   2019-05-02 18:05:14 UTC+0000
0x5c7bac70         UDPv6    :::5004                        *:*                                   2464     wmpnetwk.exe   2019-05-02 18:05:14 UTC+0000
0x5c7f9600         UDPv4    0.0.0.0:3702                   *:*                                   1368     svchost.exe    2019-05-02 18:03:14 UTC+0000
0x5c7f9600         UDPv6    :::3702                        *:*                                   1368     svchost.exe    2019-05-02 18:03:14 UTC+0000
0x5c44e1b0         TCPv4    0.0.0.0:5357                   0.0.0.0:0            LISTENING        4        System
0x5c44e1b0         TCPv6    :::5357                        :::0                 LISTENING        4        System
0x5c528010         TCPv4    0.0.0.0:445                    0.0.0.0:0            LISTENING        4        System
0x5c528010         TCPv6    :::445                         :::0                 LISTENING        4        System
0x5c534c60         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING        504      services.exe
0x5c534c60         TCPv6    :::49156                       :::0                 LISTENING        504      services.exe
0x5c535010         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING        504      services.exe
0x5c6de720         TCPv4    0.0.0.0:49154                  0.0.0.0:0            LISTENING        920      svchost.exe
0x5c6de720         TCPv6    :::49154                       :::0                 LISTENING        920      svchost.exe
```

> Answer - UDP:5005

Question 2 - Vads tag and execute protection are strong indicators of malicious processes; can you find which they are? (ANSWER format: Pid1;Pid2;Pid3)

```
python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 malfind
```

What this command does

The `malfind` plugin in Volatility Framework searches for suspicious memory regions that may contain:

- Injected code
- Shellcode
- Malware payloads
- Process injection artifacts

It specifically looks for:

- Hidden executable memory

- Memory pages with execute permissions
- Suspicious VAD (Virtual Address Descriptor) entries

This is one of the strongest indicators of malware activity during memory forensics.

```
0x0000000003ee0010 0000              ADD [EAX], AL
0x0000000003ee0012 0000              ADD [EAX], AL
0x0000000003ee0014 0000              ADD [EAX], AL
0x0000000003ee0016 0000              ADD [EAX], AL
0x0000000003ee0018 0000              ADD [EAX], AL
0x0000000003ee001a 0000              ADD [EAX], AL
0x0000000003ee001c 0000              ADD [EAX], AL
0x0000000003ee001e 0000              ADD [EAX], AL
0x0000000003ee0020 0000              ADD [EAX], AL
0x0000000003ee0022 ee                OUT DX, AL
0x0000000003ee0023 0300              ADD EAX, [EAX]
0x0000000003ee0025 0000              ADD [EAX], AL
0x0000000003ee0027 0000              ADD [EAX], AL
0x0000000003ee0029 0000              ADD [EAX], AL
0x0000000003ee002b 0000              ADD [EAX], AL
0x0000000003ee002d 0000              ADD [EAX], AL
0x0000000003ee002f 0000              ADD [EAX], AL
0x0000000003ee0031 0000              ADD [EAX], AL
0x0000000003ee0033 0000              ADD [EAX], AL
0x0000000003ee0035 0000              ADD [EAX], AL
0x0000000003ee0037 0000              ADD [EAX], AL
0x0000000003ee0039 0000              ADD [EAX], AL
0x0000000003ee003b 0000              ADD [EAX], AL
0x0000000003ee003d 0000              ADD [EAX], AL
0x0000000003ee003f 00                DB 0x0

Process: explorer.exe Pid: 1860 Address: 0x3f90000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 2, PrivateMemory: 1, Protection: 6

0x0000000003f90000  00 00 00 00 00 00 00 00 4b 5b b2 8d 2d d2 00 01   ........K[..-...
0x0000000003f90010  ee ff ee ff 00 00 00 00 28 01 f9 03 00 00 00 00   ........(.......
0x0000000003f90020  28 01 f9 03 00 00 00 00 00 00 f9 03 00 00 00 00   (...............
```

> Answer - 1860;1820;2464

Task 3 Final - IOC SAGA

Question 1 - 'www.go****.ru' (write full url without any quotation marks)

```
strings victim.raw | grep '\<www\.go....\.ru\>'
```

Question 2 - 'www.i****.com' (write full url without any quotation marks)

```
strings victim.raw | grep '\<www\.i....\.com\>'
```

Question 3 - 'www.ic****.com'**

```
strings victim.raw | grep '\<www\.ic......\.com\>'
```

Question 4 - 202..*233.* (Write full IP)

```
strings victim.raw | grep '\<202\....\.233\....\>'
```

Question 5 - *.200..164 (Write full IP)

```
strings victim.raw | grep '\<…/.200\…\.164\>'
```

Question 6 - 209.190..

```
strings victim.raw | grep '\<209\.190\....\....\>'
```

```
┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$ strings victim.raw | grep '\<www\.go....\.ru\>'
www.google.ru
www.go4win.ru
www.gocaps.ru
www.goporn.ru
^C

┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$ strings victim.raw | grep '\<www\.i....\.com\>'
http://www.iciba.com/search?s=%si
www.ikaka.com
^C

┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$ strings victim.raw | grep '\<www\.ic......\.com\>'
www.icsalabs.com
^C

┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$ strings victim.raw | grep '\<202\....\.233\....\>'
202.107.233.211
^C

┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$  strings victim.raw | grep '\<...\.200\...\.164\>'
phttp://209.200.12.164/drm/provider_license_v7.php
^C

┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$  strings victim.raw | grep '\<209\.190\....\....\>'
`http://209.190.122.186/drm/license-savenow.asp
```

The `strings` utility extracts readable text from the raw memory dump, while `grep` filters specific patterns from the output.

This technique is commonly used during DFIR investigations to quickly identify:

- Domains
- URLs
- IP addresses
- Malware artifacts
- Suspicious strings hidden inside memory

Question 7 - What is the unique environmental variable of PID 2464?

```python
python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 envars -p 2464
```

```
┌──(kav3nirmal141㉿kali)-[~/Tryhackme/Forensics /Task 3]
└─$ python2.7 ~/volatility/vol.py -f victim.raw --profile=Win7SP1x64 envars -p 2464
Volatility Foundation Volatility Framework 2.6.1
Pid      Process              Block              Variable                        Value
-------- -------------------- ------------------ ------------------------------- -----
    2464 wmpnetwk.exe         0x00000000002c47a0 ALLUSERSPROFILE                C:\ProgramData
    2464 wmpnetwk.exe         0x00000000002c47a0 APPDATA                        C:\Windows\ServiceProfiles\NetworkService\AppData\Roamin
g
    2464 wmpnetwk.exe         0x00000000002c47a0 CommonProgramFiles             C:\Program Files\Common Files
    2464 wmpnetwk.exe         0x00000000002c47a0 CommonProgramFiles(x86)        C:\Program Files (x86)\Common Files
    2464 wmpnetwk.exe         0x00000000002c47a0 CommonProgramW6432             C:\Program Files\Common Files
    2464 wmpnetwk.exe         0x00000000002c47a0 COMPUTERNAME                   VICTIM-PC
    2464 wmpnetwk.exe         0x00000000002c47a0 ComSpec                        C:\Windows\system32\cmd.exe
    2464 wmpnetwk.exe         0x00000000002c47a0 FP_NO_HOST_CHECK               NO
    2464 wmpnetwk.exe         0x00000000002c47a0 LOCALAPPDATA                   C:\Windows\ServiceProfiles\NetworkService\AppData\Local
    2464 wmpnetwk.exe         0x00000000002c47a0 NUMBER_OF_PROCESSORS           1
    2464 wmpnetwk.exe         0x00000000002c47a0 OANOCACHE                      1
    2464 wmpnetwk.exe         0x00000000002c47a0 OS                             Windows_NT
    2464 wmpnetwk.exe         0x00000000002c47a0 Path                           C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;
C:\Windows\System32\WindowsPowerShell\v1.0\
    2464 wmpnetwk.exe         0x00000000002c47a0 PATHEXT                        .COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
    2464 wmpnetwk.exe         0x00000000002c47a0 PROCESSOR_ARCHITECTURE         AMD64
    2464 wmpnetwk.exe         0x00000000002c47a0 PROCESSOR_IDENTIFIER           Intel64 Family 6 Model 42 Stepping 7, GenuineIntel
    2464 wmpnetwk.exe         0x00000000002c47a0 PROCESSOR_LEVEL                6
    2464 wmpnetwk.exe         0x00000000002c47a0 PROCESSOR_REVISION             2a07
    2464 wmpnetwk.exe         0x00000000002c47a0 ProgramData                    C:\ProgramData
    2464 wmpnetwk.exe         0x00000000002c47a0 ProgramFiles                   C:\Program Files
    2464 wmpnetwk.exe         0x00000000002c47a0 ProgramFiles(x86)              C:\Program Files (x86)
    2464 wmpnetwk.exe         0x00000000002c47a0 ProgramW6432                   C:\Program Files
    2464 wmpnetwk.exe         0x00000000002c47a0 PSModulePath                   C:\Windows\system32\WindowsPowerShell\v1.0\Modules\
    2464 wmpnetwk.exe         0x00000000002c47a0 PUBLIC                         C:\Users\Public
    2464 wmpnetwk.exe         0x00000000002c47a0 SystemDrive                    C:
    2464 wmpnetwk.exe         0x00000000002c47a0 SystemRoot                     C:\Windows
```

What this command does

The `envars` plugin in Volatility Framework extracts environment variables associated with a process.

Environment variables can reveal:

- Execution context
- User profile information
- Malware-specific variables
- Sandbox or VM artifacts
- Custom attacker configurations

> Answer - OANOCACHE

Share your win with your peers
Your progress can inspire others in your community.
Forensics completed!
kave3nirmal141 completed another room on their cyber security journey.
Completed tasks
3
Points earned
360
Streak
160
Share on social media
Share your win before starting the next room
LinkedIn
WhatsApp
Telegram
Twitter / X
Facebook
Reddit
Go to dashboard
e mouse pointer outside or press Ctrl+Alt.